

Qianqian Li

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Education

Northeastern University, Shenyang, China[🔗]

Sept 2023 – Jun 2027

Bachelor of Artificial Intelligence, School of Computer Science and Engineering

- **GPA:** 88.4/100 (Overall), 92/100 (Major/Junior)
- **Honors:** Project 985, Project 211, Double First-Class University

Nanyang Technological University (NTU), Singapore[🔗]

Aug 2024

Visiting Student, Artificial Intelligence Summer Program

Fully-funded visiting program focused on Artificial Intelligence. Engaged in intensive daily coursework and delivered a comprehensive final project presentation.

Key Courses: Machine Learning (97), Computer Graphics (94), Discrete Math (92), Computer Systems (93), Compiler Construction (91), Numerical Analysis (91), Linear Algebra (93), Image Processing (95)

Research Interests

My research centers on **3D Spatial Understanding, Environmental Perception, and Cognitive AI**. I aim to bridge physical and digital worlds through intelligent AR environments and develop **Cognitive Digital Twins** to preserve expert human reasoning—enabling AI to push the boundaries of complex scientific discovery.

Publications

A time–frequency graph fusion framework for Major Depressive Disorder diagnosis in multi-site rsfMRI data[🔗]

Journal of Affective Disorders, 2026

Co-author

Research Experience

University of North Carolina at Chapel Hill – Research Intern

Aug 2025 – Present, Remote

Advisor: Prof. Mingyu Ding[🔗]

- Developed a scalable simulation pipeline for generating 3D articulated objects with accurate physics configurations, facilitating downstream robotic manipulation research.
- Parsed MuJoCo XML files for rigid body assembly and synthesized 3D physical assets via Rodin to construct robust and diverse spatial environments.
- Migrated and refactored the core physics framework from iOS/Swift to Unity, resolving complex multi-collider interactions and stabilizing the simulation engine.

NEUDM, Northeastern University, China[🔗] – Research Assistant

Oct 2024 – Present

Advisor: Prof. Xiaocui Yang[🔗]

- Investigated and enhanced the reasoning capabilities of small multimodal models for complex sentiment analysis on Twitter 2015 and 2017 benchmark datasets.
- Explored knowledge distillation methodologies from Large Language Models (LLMs) and applied random-masking reasoning strategies to optimize model architectures and generalization.

Awards & Honors

- **National "Good" Project Rating** (Core Technical Member) Mar 2025
- **Yao Tianshun Scholarship** (Top 0.5%, 3/600 students) Oct 2024
- **Second-Class University Scholarship** (Top 10%) Sept 2024
- **Global Semifinals Excellence Award**, Zhongguancun Talents Maker Contest 2024

Selected Academic & Engineering Projects

Intelligent AR Companion: Interactive 3D Avatar in Real-world Environments

Nov 2025 – Mar 2026

- **Virtual Vision & Reconstruction:** Formulated a virtual vision pipeline employing 3D Gaussian Splatting from mobile imagery, establishing precise spatial registration between 3D scenes and real-world coordinates.
- **AR Integration:** Embedded animated 3D avatars (capable of lip-sync and gaze tracking) into an iOS framework, leveraging ARKit to handle robust spatial occlusion and continuous rendering.
- **Cognitive Systems:** Constructed a retrieval-augmented generation (RAG) backend with Chroma DB, integrating the DeepSeek API for complex dialogue and CosyVoice 2.0 for high-fidelity timbre cloning.

Autonomous Multimodal Robot Control System

Jan 2025 – Jun 2025

- Designed an autonomous multimodal agent architecture to process correlated visual and auditory sensor inputs via an asynchronous communication backend.
- Realized an AI-driven decision loop utilizing multimodal analysis models to process live video feeds, explicitly framing dynamic navigation and path-planning parameters.
- Evaluated system latency and throughput, mitigating hardware bottlenecks through the optimization of event-driven asynchronous task streaming.

Algorithm Implementation: Huffman Compression Service

Dec 2024

- Developed a networked compression platform driven by a custom C++ HTTP server (utilizing macOS native socket libraries) and an interactive web interface.
- Implemented and verified the correctness of Huffman coding algorithms for lossless data compression, processing raw text, binaries, and imaging formats up to 10MB.

Industry Experience

Lion Rock AI Lab, China Merchants Group – Robotics Intern

Jan 2026 – Mar 2026

Mentor: Zhelin Lan

- Conducted literature reviews on target-following algorithms under extreme occlusions and facilitated their translation to real-world industrial deployments.
- Maintained Linux-based robotic setups (Ubuntu, Docker, ROS) to execute map generation via advanced SLAM algorithms.

MOFA (Shenyang) Info Tech Co., Ltd. (X-Friend) – Technical Member

Sept 2024 – Present

Mentors: Prof. Zhiliang Zhu, Hongyan Lv

- Handled empirical prompt tuning and constraint formulations for generative vision models (DALL-E 3, Stable Diffusion) to solve character domain distortion artifacts in human-to-avatar tasks.

Technical Skills

Programming Languages: Python, C/C++, Swift, HTML/CSS, JavaScript, LaTeX

Frameworks & Tools: PyTorch, Unity, ROS, MuJoCo, SLAM, Docker, ARKit, RAG, Chroma, OpenCV, Git

Extracurricular Activities

Student Association for Science and Technology – Dept. Head

Sept 2024 – Jul 2025

- Organized two major campus-wide scientific dissemination events involving over 500 participants, fostering undergraduate engagement in STEM.